



PAK-AUSTRIA FACHHOCHSCHULE

INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY



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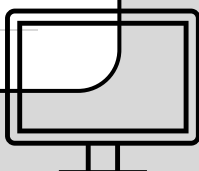
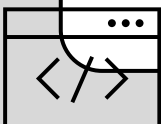
Program : Cybersecurity

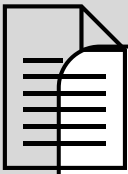
Instructor's name: Hassan Ali

Date Submitted : 10 / 07 / 26

Deadline : 10 / 07 / 26

Assignment Report # 4





• INTRODUCTION •

In this task we learned the capture and analysis of live network traffic using Wireshark, a open-source packet analyser. The main objective was to start a live capture then apply filters on them. We were given an assignment at the end; it goes as follows.

Assignments Tasks:

Task : *Traffic Analysis*, capture some Network traffic and apply some filters.

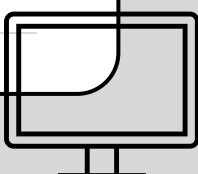
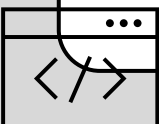
- Install and Run Wireshark.
- Start a live Capture and gather some Traffic.
- Explain and apply various filters on the traffic.

• TASK •

Task # 01 : *Traffic Analysis*, capture some Network traffic and apply some filters.

Steps Followed:

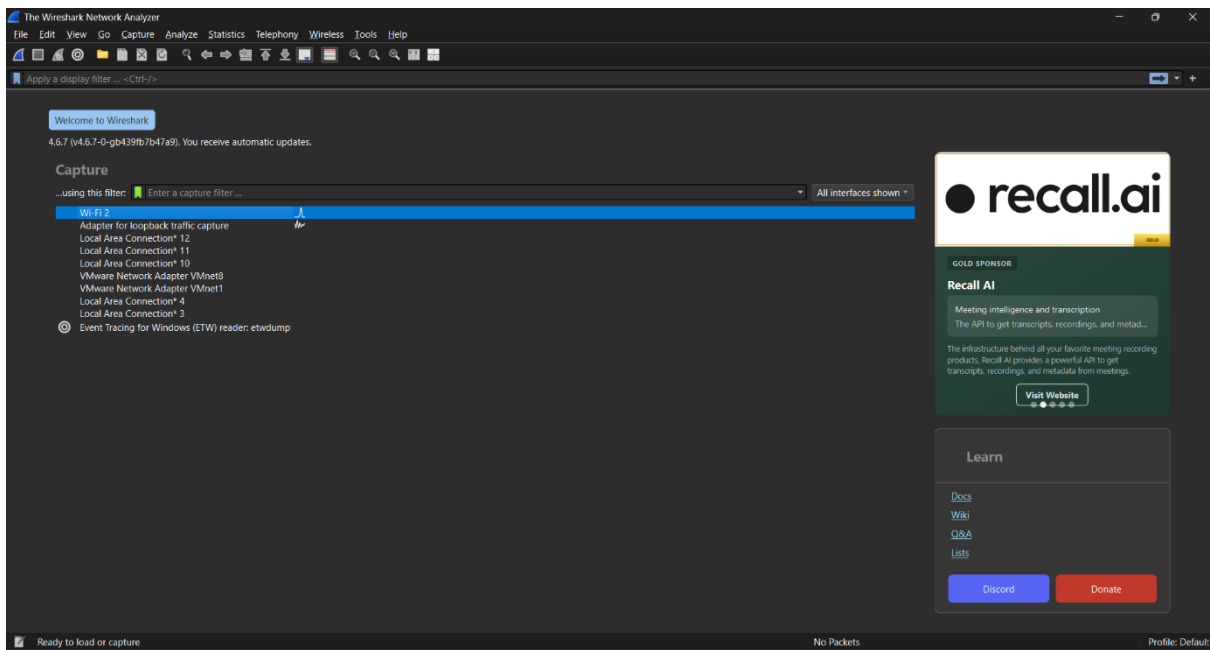
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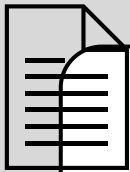
• STEP # 01 •

Step # 01 : Install and Run Wireshark.

- Install the Wireshark installer from its official website
<https://www.wireshark.org/download.html>
- Continue with installation as normal.
- Following is what the main interface looks like.



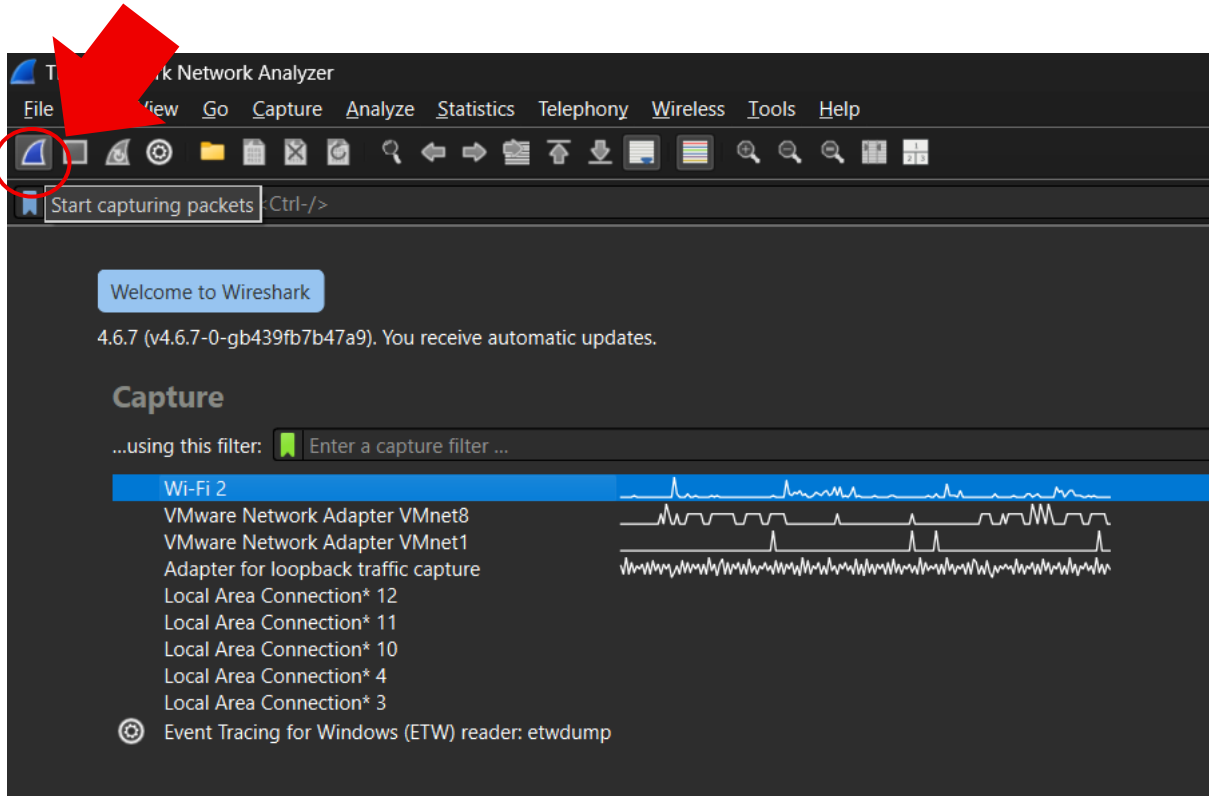
Step #01: Main Interface of Wireshark



• STEP # 02 •

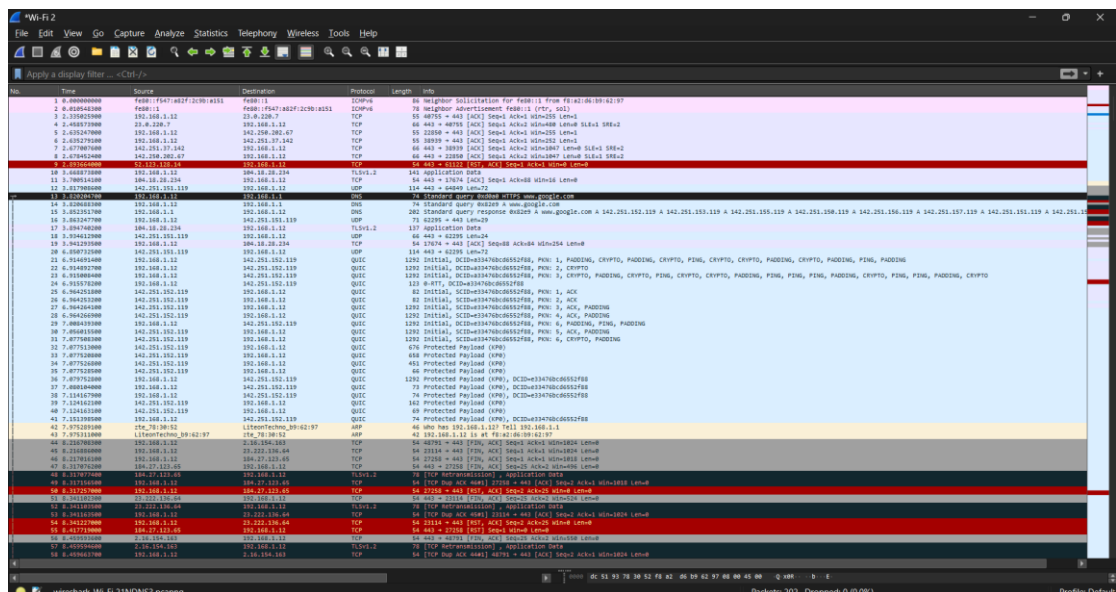
Step # 02 : Start a live Capture and gather some Traffic.

- Click on the shark fin icon to start a live capture.

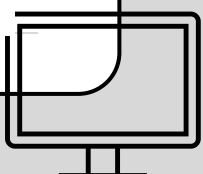
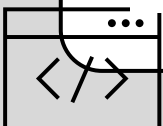


Step #02: Capturing Packets

This is what the capture looks like,



Step #02: Packets Captured

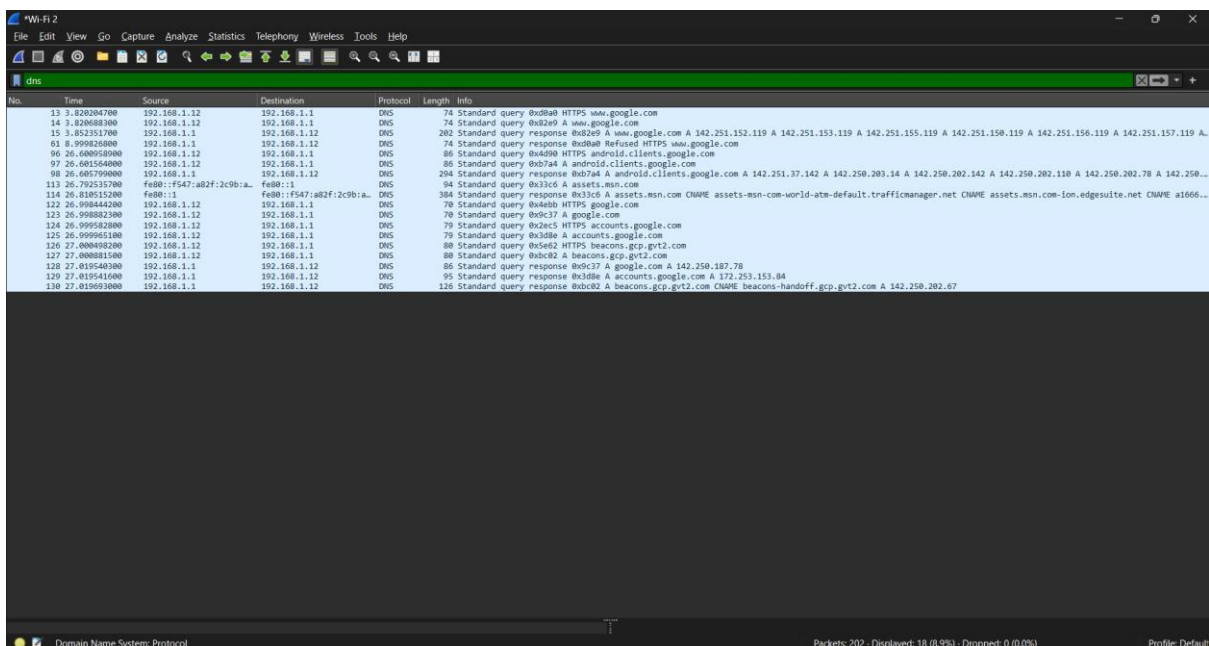


• STEP # 03 •

Step # 03 : Explain and apply various filters on the traffic.

- **DNS Filter :** Displays only DNS (Domain Name System) queries and responses.
Useful for: Seeing domain name resolutions.
- **HTTP Filter :** Shows only HTTP traffic.
Useful for: Analysing plain web traffic and requests.
- **tcp.port == 443 :** Filters HTTPS traffic.
Useful for: Viewing secure web browsing traffic.
- **TCP Filter :** Shows all TCP packets.
Useful for: General analysis of reliable connections.
- **ip.addr == 192.168.1.1 :** Shows all packets whose destination or source is 192.168.1.1.

After Applying DNS Filter,

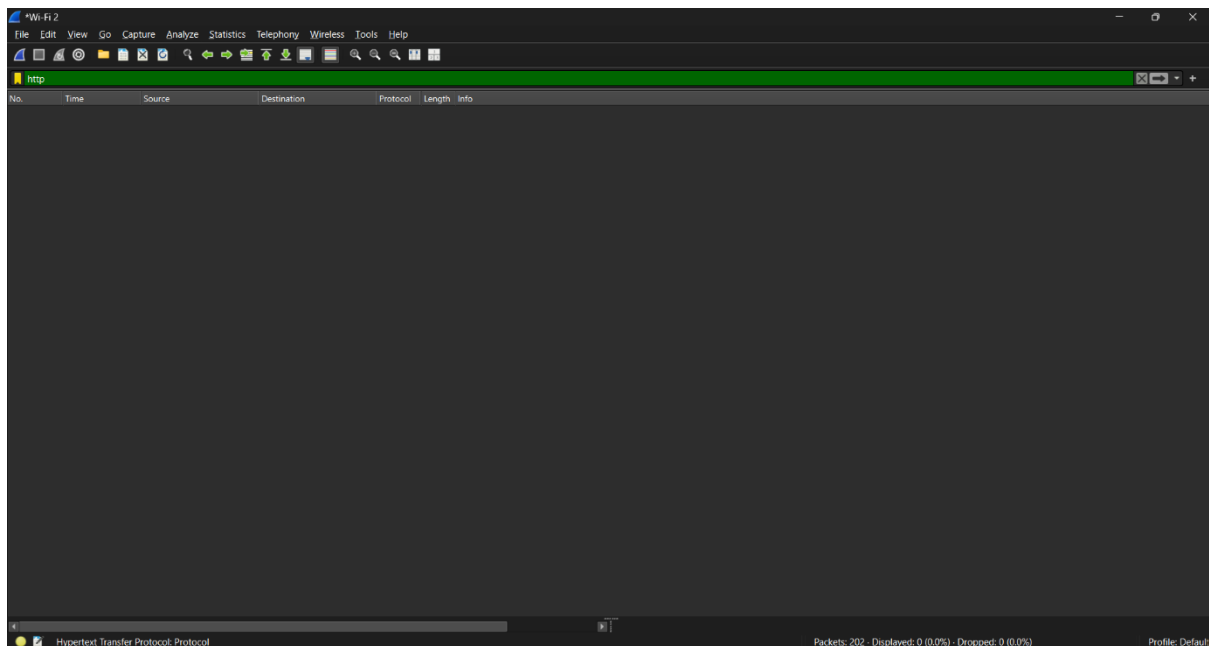


No.	Time	Source	Destination	Protocol	Length	Info
13	3.820304700	192.168.1.12	192.168.1.1	DNS	74	Standard query 0x8b98 HTTPS www.google.com
14	3.820688300	192.168.1.12	192.168.1.1	DNS	74	Standard query response 0x8b98 A www.google.com
15	3.852351700	192.168.1.1	192.168.1.12	DNS	282	Standard query response 0x8b98 A www.google.com A 142.251.152.119 A 142.251.153.119 A 142.251.155.119 A 142.251.156.119 A 142.251.157.119 A...
61	8.999626000	192.168.1.1	192.168.1.12	DNS	74	Standard query response 0x8b98 Refused HTTPS www.google.com
96	26.680958900	192.168.1.12	192.168.1.1	DNS	88	Standard query 0x4d90 HTTPS android.clients.google.com
97	26.681564800	192.168.1.12	192.168.1.1	DNS	88	Standard query response 0x4d90 A android.clients.google.com
98	26.687799000	192.168.1.1	192.168.1.12	DNS	284	Standard query response 0x4d90 A android.clients.google.com A 142.251.37.142 A 142.250.203.14 A 142.250.202.142 A 142.250.202.110 A 142.250.202.78 A 142.250...
113	26.792535700	fe80::f547:a82f:2c9b:a... fe80::1	fe80::1	DNS	94	Standard query 0x33c6 A assets.msn.com
114	26.810515200	fe80::1	fe80::f547:a82f:2c9b:a... fe80::1	DNS	384	Standard query response 0x33c6 A assets.msn.com CNAME assets.msn-com-world-atm-default-trafficmanager.net CNAME assets.msn-com-ion.edgesuite.net CNAME a1666...
122	26.958442800	192.168.1.12	192.168.1.1	DNS	78	Standard query 0x4d9b HTTPS google.com
123	26.998882300	192.168.1.12	192.168.1.1	DNS	70	Standard query 0x9c37 A google.com
124	26.999562000	192.168.1.12	192.168.1.1	DNS	79	Standard query 0x2ac5 HTTPS accounts.google.com
125	26.999651800	192.168.1.12	192.168.1.1	DNS	79	Standard query 0x3d8e A accounts.google.com
126	27.000498200	192.168.1.12	192.168.1.1	DNS	88	Standard query 0x5e62 HTTPS beacons.gcp.gvt2.com
127	27.000881500	192.168.1.12	192.168.1.1	DNS	88	Standard query 0xb482 A beacons.gcp.gvt2.com
128	27.019548300	192.168.1.1	192.168.1.12	DNS	88	Standard query response 0x9c37 A google.com A 142.250.187.78
129	27.019541600	192.168.1.1	192.168.1.12	DNS	95	Standard query response 0x3d8e A accounts.google.com A 172.253.153.84
130	27.019653800	192.168.1.1	192.168.1.12	DNS	126	Standard query response 0xb482 A beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com A 142.250.202.67

Domain Name System: Protocol Packets: 202 · Displayed: 18 (8.9%) · Dropped: 0 (0.0%) Profile: Default

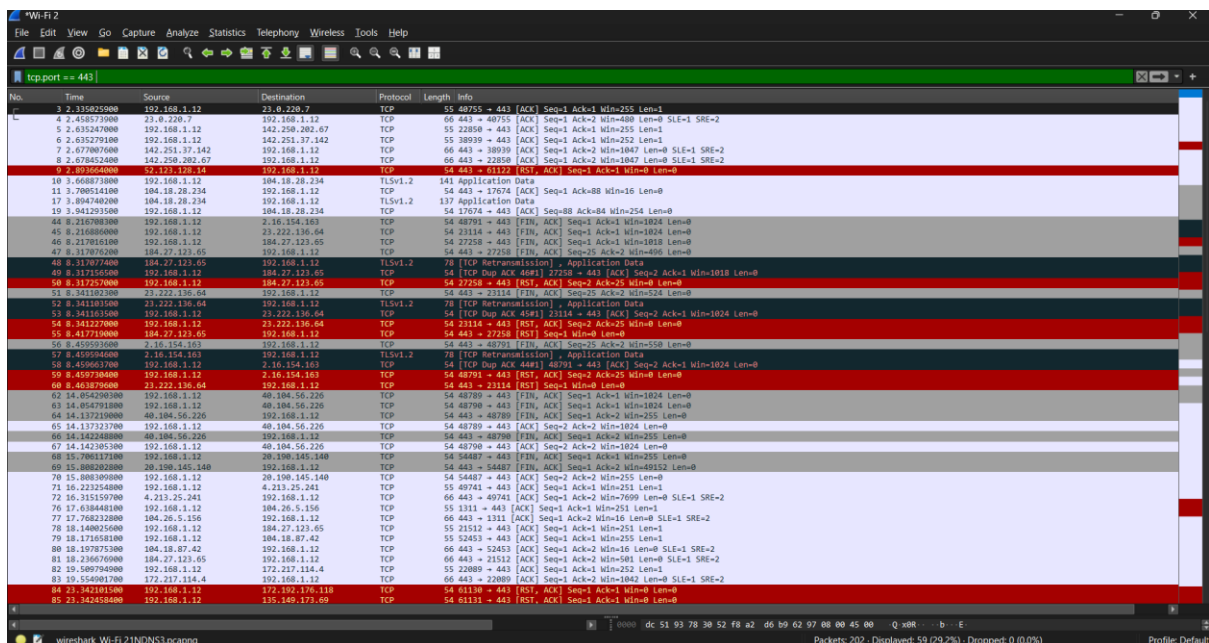
Step #03: DNS Filter

After Applying **HTTP** Filter, as we can see no http requests were recorded.



Step #03: HTTP Filter

After Applying **tcp.port == 443** Filter,



Step #03: tcp.port==443 Filter



Assignment Conclusion:

In this assignment we ran some display filters over some Network Traffic in Wireshark to get a understanding of networking fundamentals. This increased and polished my ability to use and learn more about cybersecurity.

• The End •

